

UNITED RUBBER INDUSTRIES — UNIT 1

INDUSTRIAL AUTOMATION & PROCESS ENGINEERING DIVISION

TECHNICAL PROJECT REPORT & SPECIFICATION

GUNNY BAG WEIGHING & INVENTORY MANAGEMENT SYSTEM

A complete industrial hardware-integrated automation platform featuring real-time Modbus RTU serial scale interfacing, 5-second rolling min/max stability verification, 203-DPI TSPL thermal label generation, automated audit logging, and QR-driven compound inventory tracking.

PROJECT METADATA

Organization:	United Rubber Industries (Unit 1)
System Version:	v1.0.0 (Production Release)
Software Stack:	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4
Database Layer:	PostgreSQL 14+ with Drizzle ORM (Auto-Migrating)
Hardware Interfaces:	Modbus RTU over RS-485 Serial & TSC Thermal TCP Port 9100
Thermal Label Spec:	100mm × 50mm (800 × 400 dots, 203 DPI, Model 2 QR Code)
Release Date:	September 2026

CERTIFICATE OF COMPLETION

This is to formally certify that the industrial software automation project entitled 'Gunny Bag Weighing & Compound Inventory Management System' has been successfully designed, engineered, integrated, validated, and commissioned for United Rubber Industries (Unit 1).

The system fulfills all defined operational requirements, including automated Modbus RTU serial scale interfacing, real-time rolling-window weight stability verification (spread ≤ 0.005 kg for 5 consecutive seconds), high-speed 203-DPI TSPL thermal label printing, comprehensive weighing audit logging, and QR-code enabled compound inventory tracking (Inward Receipt, CIS Staging, and Outward Dispatch).

The software architecture, database integrity mechanisms, and industrial hardware drivers have been thoroughly tested and validated on the factory floor, demonstrating high operational reliability and performance.

Project Lead / Lead Engineer

Industrial Automation Division

Plant Head / Technical Director

United Rubber Industries (Unit 1)

DECLARATION

I hereby declare that the technical specifications, architectural diagrams, hardware interfacing protocols, database schemas, and codebase documented in this report represent authentic and verified engineering work executed for United Rubber Industries (Unit 1). All operational workflows, Modbus register mappings, and thermal label templates represent active production systems.

ACKNOWLEDGEMENTS

We express our gratitude to the plant executive leadership and operations management of United Rubber Industries (Unit 1) for providing the resources, industrial scale hardware, TSC label printers, and production environment necessary to complete this project.

Special acknowledgement is given to the quality control supervisors, shop-floor operators, and warehouse managers whose practical insights shaped the high-contrast user interface, rolling-window stability filter, and seamless scan-to-advance workflows.

ABSTRACT & EXECUTIVE SUMMARY

In industrial rubber manufacturing, precise compound weighing is critical to product quality, batch consistency, and regulatory compliance. Traditional manual weighing operations—relying on manual scale reading, handwritten tags, and physical ledgers—introduce significant risks of human error, weight discrepancies, and audit gaps.

This document presents the engineering specification of the Gunny Bag Weighing System. Built with Next.js 16 (App Router), React 19, TypeScript, and PostgreSQL with Drizzle ORM, the platform unifies hardware scale reading, weight tolerance validation, thermal label printing, audit trail logging, and compound inventory lifecycle tracking into a single cohesive interface.

Hardware connectivity is achieved via a dedicated background Modbus RTU serial service (modbus-serial) that continuously samples scale registers over RS-485, coupled with a raw TCP socket client that streams TSPL commands directly to TSC thermal printers at 203 DPI. The system features a 5-second rolling min/max stability detection algorithm (WeightStability) to eliminate scale vibration noise.

EXECUTIVE SUMMARY HIGHLIGHTS

Key Features:

- Automated Modbus RTU scale communication over serial port (COM7, 9600 baud, 8N1)
- Rolling 5-second min/max weight stability verification (spread <= 0.005 kg)
- Direct TCP socket TSPL thermal printing for 100mm × 50mm labels with QR codes
- Zero-touch automated database migrations on server boot via instrumentation.ts
- Complete audit history with real-time analytics, CSV export, and PDF document generation
- Compound Inventory Lifecycle: Inward Receipt, CIS Staging, and Outward Dispatch with scan-to-advance

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CHAPTER 1: INTRODUCTION

1.1 Background & Problem Statement

United Rubber Industries (Unit 1) processes high volumes of specialized rubber compounds daily. These compounds are packaged into gunny bags with strict weight tolerances depending on customer specifications and rubber formulas. Historically, weighing was performed manually: operators visually read analog/digital scale indicators, checked weights against paper charts, hand-wrote bag tags, and recorded entries in ledgers.

This manual approach suffered from significant drawbacks: human calculation mistakes, delayed bag turnaround times, illegible tags, lack of audit traceability, vulnerability to scale drift, and data synchronization delays between weighing, staging, and dispatch departments.

1.2 Objectives & Goals

- **Automated Data Capture:** Automate real-time weight reading from industrial scales via Modbus RTU serial communication.
- **Instant Tolerance Validation:** Eliminate manual calculations by validating bag weight against dynamic item-level tolerance ranges.
- **On-Demand Thermal Printing:** Generate standardized 100mm × 50mm thermal labels with machine-scannable QR codes upon stable weigh-in.
- **Audit & Traceability:** Maintain an immutable audit trail of all weighings with operator details, timestamps, and exportable reports.
- **Inventory Control:** Provide end-to-end compound inventory tracking across Inward, CIS staging, and Outward dispatch cycles.

1.3 Project Scope & Boundaries

The system encompasses production floor operations (Master Catalog, Live Scale Interfacing, Thermal Label Printing, Historical Auditing) and raw compound warehouse logistics (Inward Receipts, Compound Issue Slips, Outward Dispatch). It is designed for on-premise local deployment on Windows workstations connected to RS-485 serial scale converters and Ethernet-connected TSC thermal printers.

CHAPTER 2: SYSTEM ARCHITECTURE & TECH STACK

2.1 High-Level Architecture

The system employs a client-server architecture running within Next.js 16. The browser client interacts with Next.js App Router API endpoints via JSON REST interfaces. Hardware communication drivers and database connection pools are maintained as singleton instances on the Node.js globalThis runtime to prevent connection thrashing and survive Hot Module Replacement (HMR).

Subsystem	Technology	Role / Implementation Detail
Frontend UI	React 19 / Tailwind CSS 4	Modern responsive operator dashboard, modals, real-time polling
Backend Server	Next.js 16 (Node.js runtime)	REST API routes, validation pipelines, TSPL socket dispatch
Database Layer	PostgreSQL 14+ / Drizzle ORM	Relational persistence, auto-migrations on startup, pool caching
Scale Interface	modbus-serial (RS-485)	Non-overlapping serial polling (1000ms), register decode
Printer Client	Node net.Socket (TCP 9100)	Raw TSPL string dispatch, 203 DPI bit-accurate QR rendering
Document Engine	PDFKit & docx	Server-side vector PDF generation and Word document packaging

2.2 Key Architectural Decisions

- Direct Socket Printing:** No OS printer driver or Windows spooler required; TSPL commands are sent directly via TCP port 9100.
- Zero-Touch Migration:** Database schema is validated and migrated automatically on application boot via instrumentation.ts.
- Non-Overlapping Polling:** Scale polling uses an async lock to prevent queue buildup during serial communication latency.
- Denormalized Audit Trail:** Part number and description are copied into history records to preserve historical integrity.

CHAPTER 3: REQUIREMENTS SPECIFICATIONS

3.1 Functional Requirements Matrix

Req ID	Module	Description & Acceptance Criteria
FR-01	Master Parts	Maintain master part catalog with min/max weight per piece and pieces per bag.
FR-02	Live Scale	Poll Modbus RTU scale over COM port at 1000ms intervals and stream weight to UI.
FR-03	Tolerance Engine	Compute bag bounds (min × qty, max × qty) and classify status (OK/Under/Over).
FR-04	Stability Filter	Detect 5-second weight stability (spread <= 0.005 kg) before auto-recording.
FR-05	Thermal Print	Generate TSPL commands for 100×50mm labels and dispatch to TSC printer.
FR-06	Weighing History	Persist all weighing events; provide search, date filter, CSV export, and PDF.
FR-07	Compound Inward	Record raw material inward, compute supplier batching, and print inward label.
FR-08	Compound CIS	Issue compound against verified inward batches with remaining balance checks.
FR-09	Compound Outward	Dispatch finished compound from CIS stage and generate dispatch documentation.

3.2 Non-Functional Requirements

- **Low Latency UI:** Weight display updates every 500ms; Modbus read latency under 100ms.
- **Timezone Standardization:** All timestamps persisted in UTC and rendered in Asia/Kolkata (IST, UTC+5:30).
- **Fault Tolerance:** Automatic mock scale fallback when physical hardware is detached during testing.
- **Scanner Compatibility:** Thermal label QR code formatted to Model 2, ECC M for industrial scanner compatibility.

CHAPTER 4: DATABASE SCHEMA & DATA INTEGRITY

4.1 Relational Data Model

The relational database schema is modeled in PostgreSQL using Drizzle ORM. It balances normalization for catalog entities with historical denormalization for weighing transactions.

4.2 Table Specifications

Table Name	Key Columns	Constraints & Data Types
parts	id, part_number, description, min_weight, max_weight, quantity	part_number UNIQUE VARCHAR(100), NUMERIC(10,3), INT >= 1
weighing_history	id, part_id, part_number, actual_weight, quantity, status, operator_name, recorded_at	part_id FK -> parts.id ON DELETE CASCADE, status VARCHAR(20)
compound_inwards	id, inward_number, part_id, quantity, supplier, batch_number, label_code, received_at	inward_number UNIQUE, part_id FK, label_code UNIQUE
compound_cis	id, cis_number, inward_id, part_id, quantity, operator_name, label_code, created_at	cis_number UNIQUE, inward_id FK, label_code UNIQUE
compound_outwards	id, outward_number, cis_id, part_id, quantity, destination, label_code, dispatched_at	outward_number UNIQUE, cis_id FK, label_code UNIQUE

4.3 Status Classification Logic

The bag status is evaluated strictly on the backend and verified on the client using the following mathematical bounds:

TOLERANCE CLASSIFICATION ALGORITHM

Minimum Allowed Bag Weight = min_weight × quantity
Maximum Allowed Bag Weight = max_weight × quantity

if (actual_weight < Minimum Allowed Bag Weight) != UNDERWEIGHT
else if (actual_weight > Maximum Allowed Bag Weight) != OVERWEIGHT
else != OK (Accepted)

CHAPTER 5: DETAILED MODULE IMPLEMENTATION

5.1 Master Catalog & Bag Configuration

The Master Catalog module allows plant administrators to register rubber compound parts. For each part, the system stores per-unit min/max weights in kg and standard pieces per bag. The UI provides real-time bag weight calculation previews, search filtering with a 350ms debounce, and batch label printing.

5.2 Live Scale & Rolling Min/Max Stability Algorithm

Industrial scales in manufacturing plants frequently experience vibrations from conveyors, forklifts, and mechanical presses. Standard threshold checks comparing two adjacent readings are vulnerable to slow weight drift. The Gunny Bag Weighing System implements a robust rolling-window stability algorithm:

WEIGHT STABILITY FILTER

Algorithm Parameters:

- STABLE_DURATION_MS = 5000 ms
- WEIGHT_TOLERANCE_KG = 0.005 kg
- MAX_READING_AGE_MS = 2000 ms

Evaluation Logic:

1. On each tick, add reading (weight, timestamp) to rolling buffer.
2. Evict samples older than 5,000 ms.
3. Compute spread = max(buffer.weights) - min(buffer.weights).
4. If spread $\leq$ 0.005 kg AND buffer span $\geq$ 5000 ms -> Report STABLE.
5. Upon auto-print, system locks until scale drops to $\leq$ 0.005 kg (rearm cycle).

5.3 TSPL 203-DPI Thermal Label Design

Thermal printing is implemented using the Thermal Smart Printer Language (TSPL) targeting standard 100mm × 50mm label stock at 203 dots per inch (800 × 400 dots total canvas). The layout includes a bold company header, outer bounding box, vertical divider bar, native matrix QR code generated via qrcode, and crisp aligned data rows using Font 2.

5.4 Compound Inventory Subsystems

The Compound Inventory module introduces end-to-end multi-stage tracking: Inward Receipt (raw materials received from external suppliers), Compound Issue Slip / CIS (materials staged for internal processing), and Outward Dispatch (finished goods ready for shipment). Each stage generates a unique reference number, prints a 100×50mm label, and provides a scan-to-advance transition.

CHAPTER 6: TESTING & QUALITY ASSURANCE

6.1 Testing Methodology

Comprehensive verification was executed across hardware drivers, business logic algorithms, database persistence, and user interfaces using both simulated mocks and physical shop-floor hardware.

Test Case ID	Module	Test Scenario	Expected Outcome	Status
TC-01	Modbus Serial	Disconnect serial cable mid-operation	Service logs warning, switches to mock without crash	PASSED
TC-02	Stability Engine	Introduce sinusoidal weight jitter of 0.02kg	Stability flag remains FALSE; no auto-print triggers	PASSED
TC-03	Stability Engine	Steady weight held for 5.1 seconds	Auto-saves record and dispatches print socket	PASSED
TC-04	Tolerance Logic	Part PN-01 (100 pcs, 0.5-1.0kg) bag at 48.0kg	Classified as UNDERWEIGHT with amber warning banner	PASSED
TC-05	Printer Client	Printer offline / IP unreachable	Socket returns timeout in 5000ms; error displayed in UI	PASSED
TC-06	Inventory Cycle	Scan Inward QR code in CIS view	Automatically opens pre-filled CIS modal for batch	PASSED

CHAPTER 7: DEPLOYMENT, SOP & FUTURE SCOPE

7.1 Standard Operating Procedure (SOP) for Operators

- **1. Startup:** Power on the weighing scale indicator and TSC label printer. Ensure green network LED on printer.
- **2. Open Weighing:** Open browser and navigate to <http://localhost:3000/weighing>. Verify green scale indicator.
- **3. Select Part:** Click 'Select Part' and choose the target rubber compound part number.
- **4. Place Bag:** Place the filled gunny bag onto the scale platform. Ensure bag does not touch side rails.
- **5. Weighing:** Observe weight display. Status will turn green (OK). Hold stable for 5 seconds.
- **6. Labeling:** Label automatically prints from TSC printer. Affix label to the center of the gunny bag.
- **7. Rearm:** Remove bag from scale platform. System rearms automatically once weight reaches zero.

7.2 Production Environment Configuration

Configuration is managed via the .env file. The production setup defines the database connection URL, printer IP address (192.168.1.75:9100), serial port (COM7, 9600 baud, 8N1), slave ID 1, function code 4, register 0, and divisor 100.

7.3 Future Enhancements

- **Enterprise ERP Sync:** Integration with cloud ERP platforms (SAP / Oracle) for automated purchase order reconciliation.
- **Multi-Scale Topology:** Support for multiple weighing scale stations communicating concurrently to a central server.
- **RFID Asset Tracking:** Introduction of industrial RFID tags for automated bag tracking across warehouse zones.

CHAPTER 8: REFERENCES

1. Modbus Organization, 'Modbus Application Protocol Specification V1.1b3', 2012.
 2. TSC Auto ID Technology Co., Ltd., 'TSPL/TSPL2 Programming Language Reference Manual', 2021.
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 5. ISO/IEC 18004:2015, 'Information technology — Automatic identification and data capture techniques — QR Code bar code symbology specification'.
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